1. The Von Neumann Architecture

Most modern computer systems are based on the **Von Neumann model**, which introduced the concept of the **Stored Program**.

- **Components:**
  - **CPU (Central Processing Unit):** The "brain" of the system.
  - **Main Memory (RAM):** Stores both data and instructions.
  - **I/O System:** Interfaces for input and output.
  - **System Bus:** The communication pathway (Data, Address, and Control buses).

Key Concept: In this architecture, instructions and data share the same memory space and the same bus system.

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2. Detailed Processor Organization

The CPU is the primary execution unit. Its internal structure includes:

- **Control Unit (CU):** Decodes instructions and generates control signals to synchronize hardware.
- **Arithmetic Logic Unit (ALU):** Performs all arithmetic (e.g., A+B) and logical (e.g., A>B) operations.
- **Registers:** High-speed storage locations within the CPU.
  - **Program Counter (PC):** Holds the address of the *next* instruction.
  - **Instruction Register (IR):** Holds the current instruction being decoded.
  - **Memory Address Register (MAR):** Holds the address of the memory location being accessed.
  - **Memory Data Register (MDR):** Holds the actual data being read from or written to memory.
  - **Accumulator (ACC):** Stores intermediate results of the ALU.

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3. The Fetch-Execute Cycle

The CPU processes instructions in a continuous cycle:

1. **Fetch:** The CPU copies the address from the **PC** to the **MAR**, then fetches the instruction from memory into the **MDR**, then moves it to the **IR**. The PC is then incremented.
2. **Decode:** The **CU** interprets the instruction in the IR.
3. **Execute:** The **CU** sends signals to the **ALU** or other components to perform the task.
4. **Store:** Results are saved back into a register or memory if necessary.

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4. Memory Hierarchy

Computer systems use different types of memory to balance **speed**, **cost**, and **capacity**.

Memory Type	Speed	Capacity	Volatility
Registers	Fastest	Smallest	Volatile
Cache Memory	Very Fast	Small	Volatile
Main Memory (RAM)	Fast	Moderate	Volatile
Secondary Storage	Slow	Large	Non-volatile

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5. System Performance Factors

The performance of a computer system is not just determined by one factor, but a combination:

- **Clock Speed:** Measured in Gigahertz (GHz); the number of cycles a CPU can perform per second.
- **Word Size:** The number of bits the CPU can process at one time (e.g., 32-bit or 64-bit).
- **Bus Width:** Determines how much data can be moved at once.
- **Number of Cores:** Multiple processing units on a single chip.
- **Cache Size:** Larger cache reduces the time the CPU spends waiting for data from RAM.

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6. Classification of Computers

- **By Size/Power:** Supercomputers, Mainframes, Minicomputers, Microcomputers (PCs, Laptops).
- **By Data Representation:** * **Digital:** Processes discrete data (0s and 1s).
    - **Analog:** Processes continuous data (voltage, pressure).
    - **Hybrid:** Combines both.

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1. The Processor's Internal Bus System

While the **System Bus** connects the CPU to RAM and I/O, the **Internal Bus** connects the ALU, Control Unit, and Registers.

- **Data Bus:** Bi-directional; carries the actual data (operands) between the CPU and memory.

- **Address Bus:** Unidirectional; carries the memory address the CPU wants to access. The width of this bus determines the **Maximum Addressable Memory** (e.g., a 32-bit address bus can access 232 bytes or 4GB).
- **Control Bus:** Bi-directional; carries signals like `READ`, `WRITE`, and `INTERRUPT` to synchronize components.

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2. Cache Memory and its Levels

Cache is a small, high-speed type of **SRAM (Static RAM)** that sits between the CPU and the slower **DRAM (Dynamic RAM/Main Memory)**. It stores frequently used instructions to reduce the "Von Neumann Bottleneck."

- **L1 Cache:** Located *inside* the CPU core. Fastest but smallest (KB).
- **L2 Cache:** Usually shared between a few cores or dedicated to one. Larger than L1.
- **L3 Cache:** Shared across all cores of the CPU. Slowest of the caches but much faster than RAM.

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3. Instruction Sets and CPU Design

AL ICT often distinguishes between two primary philosophies in CPU design:

Feature	CISC (Complex Instruction Set)	RISC (Reduced Instruction Set)
Instruction Length	Variable length	Fixed length
Complexity	Single instructions do complex tasks	Simple instructions, combined for tasks
Cycles	May take multiple cycles per instruction	Aims for one instruction per cycle
Examples	Intel x86, AMD	ARM (used in iPhones/Androids), Apple M1/M2

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4. Pipelining (Advanced Processing)

To increase performance, modern CPUs use **Pipelining**. Instead of waiting for one instruction to finish the entire Fetch-Decode-Execute cycle, the CPU starts fetching the *next* instruction while the current one is being decoded.

Analogy: Think of an assembly line. While one worker is putting the engine in a car, the worker behind them is already starting the chassis of the next car.

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5. Input/Output (I/O) Management Techniques

The CPU doesn't just "talk" to the keyboard or disk; it manages them using specific methods:

1. **Programmed I/O:** The CPU stays in a loop, constantly checking if the device is ready (Polling). This wastes CPU time.
2. **Interrupt-driven I/O:** The device sends an **Interrupt Signal** to the CPU when it's ready. The CPU pauses its current task, handles the device, and then returns.
3. **Direct Memory Access (DMA):** High-speed devices (like Hard Drives) transfer data directly to RAM without involving the CPU for every byte. This is the most efficient method for large data transfers.

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6. Logic Gates and Hardware Realization

At the most basic level, all computer processing is done via logic gates. You should be familiar with:

- **AND, OR, NOT:** The basic building blocks.
- **NAND, NOR:** Universal gates (can build any other gate).
- **XOR, XNOR:** Used in arithmetic circuits like Adders.